

Example6 stan

```

data {
  int<lower=0> n; // # time points (usually square)
  vector<0,1> y; // mean corrected values at time t
}

parameters {
  real mu; // log mean log-odds
  real<lower=0> sigma; // log standard deviation of log-odds
  real<lower=0> rho; // within mean scale factor
  real<lower=0> tau; // precision of outcome
}

vector<0,1> z; // log likelihood of time t

model {
  rho ~ normal(0.5,0.5);
  sigma ~ gamma(1.5,0.5);
  tau ~ normal(1);
  mu ~ normal(1.5,0.8);
  y ||~ multi_normal(mu, rho*sigma*tau*diag(n));
  for (t in 1:n) {
    y[t] ~ multi_normal(mu+(rho-1)*z[t], rho*sigma);
  }
  y ~ student_t(nu=2, exp2=0.2);
}

```

[illegible]

Other R packages that implement HMC

[illegible]

rstanarm
<https://mc-stan.org/users/documentation/case-studies/rstanarm.html> (<https://mc-stan.org/users/documentation/case-studies/rstanarm.html>) Introduction to multilevel modeling using rstanarm: A tutorial for education researchers Joohwi Lee (joohwi@berkeley.edu) (mailto:joohwi@berkeley.edu), Nicholas Sim, Feng Ji, and Sophia Rabe-Hesketh Monday, April 24, 2016

Package **bayesplot** (<https://mc-stan.org/bayesplot/>)

Thomas and Yajima (2021) say the following in their 2021 article Learning Hamiltonian Monte Carlo in R (<https://doi.org/10.1080/09637453.2021.2052228>):

If you are just getting started with bayesplot we recommend starting with the tutorial vignettes, the examples throughout the package documentation, and the paper Visualization in Bayesian workflow: Gabry et al. (2019) Visualization in Bayesian workflow. J. R. Stat. Soc. A, 182: 389-402. (<https://doi.org/10.1111/rssa.12378>)

Hamiltonian Monte Carlo (HMC) literature

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